LinkedIn Job Market Analysis — Project Documentation

DEPI Graduation Project · Junior Data Analysis Track

1. Project Overview

This project analyzes the LinkedIn Job Postings dataset to understand what the job market demands: which roles, skills, industries, locations, and salary levels dominate, and how work type and experience level shape opportunities. The final deliverable is a cleaned relational dataset and an interactive Power BI dashboard that answers practical questions for job seekers, recruiters, and career advisors.

Dataset snapshot

Table	Rows
postings	123,849
companies	24,473
employee_counts	35,787
salaries	40,785
job_skills	213,768
job_industries	164,808
benefits	67,943

Postings were listed between late March and mid April 2024. Every `job_id` in the postings table is unique.

Objectives

1. Identify the most in-demand job titles, skills, and industries.
2. Compare salaries by experience level, work type, industry, and location.
3. Measure how remote and full-time roles are distributed.
4. Show which companies hire the most and how engagement (views and applies) varies.
5. Deliver a clear dashboard that non-technical users can explore.

2. Project Planning

Methodology: iterative, with each phase reviewed before moving on.

Phase	Activities	Output
1. Understanding	Explore the 11 CSV files, define business questions	Data dictionary, question list
2. Cleaning	Handle nulls, fix types, remove duplicates, standardize text	Clean tables
3. Modeling	Build a star-style relational model and relationships	Data model
4. Analysis	Explore trends, compute measures	Findings
5. Visualization	Build the Power BI dashboard	.pbix report
6. Documentation	Write documentation, README, presentation	Final submission

Tools: Python (pandas) for cleaning and profiling, Power BI with DAX for modeling and visuals, GitHub for source code and version control.

Risks and mitigation

Risk	Mitigation
Salary is missing in most postings (about 71% have no pay period or currency)	Analyze salary only on the subset that has it and state the sample size on every salary visual
Large postings file (over 500 MB)	Keep only the columns needed, drop long text fields from the model
Mixed pay periods (hourly, monthly, yearly)	Use the normalized salary column or convert to a yearly basis
Location is free text	Split into city and state, group country-level entries separately

3. Stakeholder Analysis

Stakeholder	Role	Interest	Key questions
Job seekers	End users	Find realistic roles and salaries	Which skills should I learn? Where are the openings?
Recruiters and HR teams	End users	Benchmark pay and competition	What do similar companies offer? Which roles get the most applies?
Career counselors and universities	Secondary users	Guide students toward demand	Which industries and levels are hiring entry-level talent?
Companies	Data subject	Employer branding and hiring visibility	How do we compare to competitors in postings and engagement?
Project team	Builders	Deliver an accurate, usable product	Is the data clean and the insight reliable?
DEPI evaluators	Reviewers	Assess technical and analytical quality	Is the work complete, documented, and reproducible?

Priority: job seekers and recruiters are the primary audience, so the dashboard focuses on skills, salary, and location insights first.

4. Database Design

The data is organized as a relational model centered on the postings table.

4.1 Tables and keys

Table	Primary key	Foreign keys	Description
postings	job_id	company_id → companies	One row per job posting: title, work type, experience level, location, salary fields, views, applies, dates
companies	company_id	none	Company name, size, city, state, country

Table	Primary key	Foreign keys	Description
company_industries	company_id + industry	company_id → companies	Industries each company belongs to
company_specialities	company_id + speciality	company_id → companies	Company specialities
employee_counts	company_id + time_recorded	company_id → companies	Employee and follower counts over time
salaries	salary_id	job_id → postings	Min, median, max salary, pay period, currency
benefits	job_id + type	job_id → postings	Benefits offered per posting
job_skills	job_id + skill_abr	job_id → postings, skill_abr → skills	Skills required per posting
job_industries	job_id + industry_id	job_id → postings, industry_id → industries	Industries tagged per posting
industries (mapping)	industry_id	none	Industry names
skills (mapping)	skill_abr	none	Skill names

4.2 Relationships

- One company has many postings (1 : many).
- One posting has many skills, industries, and benefits (1 : many through bridge tables).
- Skills and industries are many-to-many with postings, resolved by `job_skills` and `job_industries`.
- One company has many employee-count snapshots (1 : many).

4.3 Cleaning rules

Issue	Rule
Timestamps stored as Unix milliseconds	Convert to datetime
Duplicate postings or company rows	Drop duplicates on the primary key

Issue	Rule
Missing salary, applies, remote flag	Keep as null, do not impute, flag with an indicator column
`remote_allowed` empty for most rows	Treat null as not specified, not as No
Company name missing (about 1.4%)	Keep the row, label as Unknown
Free-text location	Split into city and state, keep original
Mixed pay periods	Convert to a yearly salary for comparison

5. UI/UX Design

Design principles: simple, one message per page, consistent colors, filters on the left or top, and a sample-size note wherever data is incomplete.

Audience: non-technical users who want quick answers.

5.1 Dashboard pages

Page	Purpose	Main visuals
1. Overview	Market snapshot	KPI cards (total postings, companies, median salary, remote share), postings over time, work type donut
2. Roles and Experience	What is being hired	Top job titles bar chart, experience level distribution, work type by level
3. Skills	What to learn	Top skills bar chart, skills by industry matrix
4. Salary	Pay benchmarks	Salary by experience level, by industry, by work type, salary distribution histogram
5. Companies and Industries	Who is hiring	Top hiring companies, industries ranking, company size breakdown
6. Geography	Where the jobs are	Map or bar chart by state and city, remote versus on-site by region

5.2 Interaction

- Global slicers: work type, experience level, industry, state, date range.
- Drill-through from a job title to its skills and salary range.

- Tooltips with sample size and share of total.

5.3 Style

- Palette: one primary blue for the main series, grey for context, one accent color for highlights.
- Typography: one font family, large KPI numbers, small muted captions.
- Layout: consistent grid, same filter position on every page, navigation buttons at the top.

5.4 Accessibility

- Sufficient contrast, no color-only meaning, descriptive titles that state the takeaway.